

c 130–c 210 CE

GALEN

Greek physician Claudius Galen, who came from the city of Pergamum (now in modern Turkey), was one of the ancient world's most influential doctors. He believed in direct observation of patients, including taking their pulses. Galen saw

good health as the balanced working of all the body's organs, and was an expert anatomist.



In 250 CE, Diophantus of Alexandria was the first to use letters and symbols to show algebraic equations in his book *Arithmetica*.

The dome of Hagia Sophia is 107 ft (32.5 m) in diameter.

532–537 CE

Building of Hagia Sophia

The Byzantine Emperor Justinian asked Greek architects Anthemius and Isidore to build the church of Hagia Sophia in Constantinople (modern Istanbul). They set a round dome over a square base by using curved triangular sections of stone called pendentives, which strengthened the structure. Hagia Sophia remained the world's largest domed building for about a thousand years.

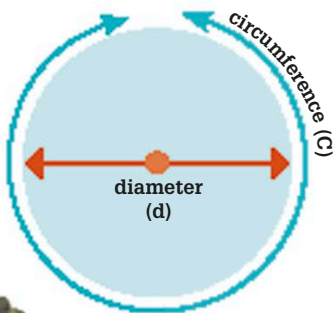
600

800

475–499 CE

Calculating pi

For hundreds of years, mathematicians had tried to calculate the value of pi (the distance around a circle, or circumference, divided by its diameter, represented by the symbol π). In about 475 CE, Chinese mathematician Zhu Chongzhi calculated pi to seven decimal places and in 499 CE, Indian mathematician Aryabhata estimated it to be 3.1416, which is correct to four decimal places.



$$\pi = C/d$$

628 CE

Negative numbers

Indian mathematician Brahmagupta was the first to set out rules for using negative numbers in calculations. These included the rule that multiplying two negative numbers gives a positive number.

750 CE

First written work on the astrolabe

The astrolabe, invented around 100 BCE, was a device with movable circles used by ancient astronomers to calculate the positions of the Sun and stars. In the 8th century, it was greatly developed by Islamic astronomers, and one of them, al-Fazari, wrote the first-ever work on the astrolabe.

Most astrolabes are portable.

Face adjusts to show appearance of sky at a given time.



Medieval Arabic brass astrolabe, dating from around 1100